



Demonstration of high-speed thin-film lithium-niobate-on-insulator optical modulators at the 2- μm wavelength

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Abstract: Optical communication wavelength is being extended from the near-infrared band of 1.31/1.55 μm to the mid-infrared band of 2 μm or beyond for satisfying the increasing demands for high-capacity long-distance data transmissions. An efficient electro-optic (EO) modulator working at 2 μm is highly desired as one of the indispensable elements for optical systems. Lithium niobate (LiNbO_3) with a large second-order nonlinear coefficient is widely used in various EO modulators. Here, we experimentally demonstrate the first Mach-Zehnder EO modulator working at 2 μm based on the emerging thin-film LiNbO_3 platform. The demonstrated device exhibits a voltage-length product of 3.67 V $\cdot\text{cm}$ and a 3-dB-bandwidth of >22 GHz which is limited by the 18 GHz response bandwidth of the photodetector available in the lab. Open eye-diagrams of the 25 Gb/s on-off keying (OOK) signals modulated by the fabricated Mach-Zehnder EO modulator is also measured experimentally with a SNR of about 14 dB.

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1. Introduction

High-speed electro-optic (EO) modulators are key components in many optical systems. Various materials have been utilized for realizing high-speed EO modulators [1–5]. Among them, lithium niobate (LN) has become one of the most attractive materials because of the wide transparency window and the large EO coefficient [6]. The waveguide of a conventional LN modulator in a bulky structure is fabricated by titanium-diffusion or proton-exchange. The refractive-index contrast is quite small in such a waveguide, which significantly increases the footprint of the device and causes a low modulation efficiency [7]. In recent years, thin-film-LN-on-insulator (LNOI) has become commercially available, which offers high refractive-index contrast and excellent compatibility with the complementary metal oxide semiconductor (CMOS) processes [8]. Currently, high-performance EO modulators on LNOI have been demonstrated for C-band ($\sim 1.55 \mu\text{m}$), showing a low drive voltage and a high modulation speed [9–12].

Recently, the 2- μm wavelength-band has become attractive for many important applications [13–16]. In particular, it is expected to be a new window for optical communication in potential. Currently, great progress has been made in the development of the EO modulators for the 2- μm wavelength-band [17–20] by utilizing silicon photonic technology. Basically, silicon EO modulators are realized by modifying the carrier concentration in silicon. However, this

modulation effect is intrinsically nonlinear, absorptive, and sensitive to the temperature fluctuation, causing the challenges for high-performance modulation. Fortunately, LiNbO_3 has very good properties for EO modulation, such as broad transparency window ranging from the visible to the mid-infrared, uniform EO coefficients in a wide wavelength spectrum, and good thermal stability [21,22]. Therefore, LNOI is expected to realize better EO modulators at the 2- μm wavelength-band than silicon.

In this paper, we demonstrated the first LNOI EO modulator at the 2- μm wavelength-band. Here the LNOI waveguide is chosen to be wider than 1 μm , so that one can use the process of UV contact lithography (instead of E-beam lithography used previously), which is helpful for the future massive production. The physical etching process of Ar^+ plasma was also optimized to achieve smooth sidewalls and thus low scattering losses. Furthermore, the waveguide structure design is also optimized carefully further for the impedance matching as well as the velocity matching. The demonstrated device exhibits an on-chip loss of around 6 dB, a voltage-length product of 3.67 V·cm, and an EO bandwidth of more than 22 GHz (limited by the bandwidth of the photodetector used in the experimental setup). The performance of the device can be further enhanced by improving the quality of the etching process and upgrading the measurement setup with a large bandwidth. The present EO modulator is promising to play a key role in 2- μm -wavelength optical systems.

2. Design and fabrication

Figure 1(a) shows the schematic configuration of the presented MZI EO-modulator based on LNOI, and Fig. 1(b) shows the schematic of the cross-section of the waveguide in the modulation region. In our design, both input and output waveguides are connected with a 90° waveguide bend in order to allow sufficient space for fibers and high-speed radio frequency (RF) probes during the EO modulation test.

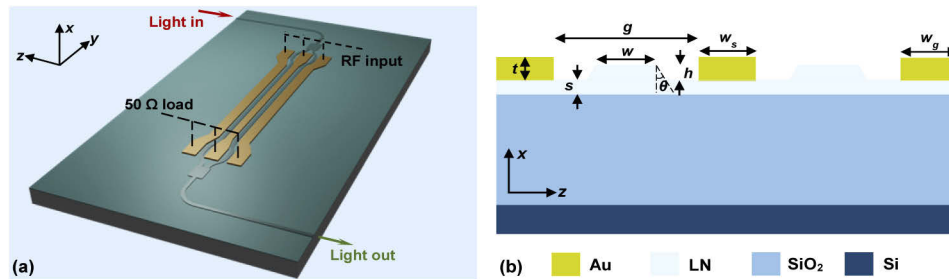


Fig. 1. (a) Schematic configuration of the presented LNOI EO-modulator; (b) Cross-section of the LNOI waveguides in the modulation region.

For light at 1.95 μm travelling in x-z plane on an x-cut 600-nm LNOI waveguide, the effective refractive indices of the TE_0 (solid) and TE_1 (dashed) modes are calculated as a function of the top width w when choosing different rib heights h , as shown in Fig. 2(a). Regarding the resolution requirement of UV lithography ($\sim 1 \mu\text{m}$), the top width w is chosen as 1.1 μm , while the rib height h should be no more than 400 nm in order to satisfy the single-mode condition. Figure 2(b) shows the simulated propagation loss caused by the metal absorption as the gap width g varies by assuming that the electrodes thickness $t = 500 \text{ nm}$. The refractive index of the metal is from [23]. It can be seen that the metal absorption decreases as the rib height h increases. Definitely, the gap width g should be large enough for any given rib height, so that the metal absorption is neglectable. For example, when the metal absorption loss is required to be less than 0.02 dB/cm (see the red dashed line), the gap width g should be larger than 8.5, 7.3, 6.5, 5.9, and 5.5 μm for the cases with different rib heights of $h = 200, 250, 300, 350$, and 400 nm,

respectively. Driven with the push-pull configuration, the voltage-length product is given as $V_{\pi}L = V\lambda_0/(4\Delta n_{eff})$, where Δn_{eff} is the effective refractive index change of the fundamental TE mode with and without the applied voltage V . Figure 2(c) shows the simulated voltage-length product $V_{\pi}L$ when choosing different gap widths g and rib heights h . Basically speaking, in the waveguide design, deeper etching results in weaker EO overlap integral, while shallower etching results in higher metal loss and thus larger electrodes gap is required. Thus, there is a trade-off when choosing the etching depth. As shown in Fig. 2(c), there is a minimal value of $V_{\pi}L$ achieved as 4.06 V·cm when choosing $g = 6.5 \mu\text{m}$ and $h = 300 \text{ nm}$. Figure 2(d) shows the simulated TE₀ mode in the modulated LNOI waveguides as well as the electric field for a given voltage, and the inset shows the RF field profile at the frequency of $f = 100 \text{ GHz}$. One can see that there is an overlap between the optical and electric fields, which is similar to the results reported previously. As a short summary, the parameters of the designed waveguide are given as $w = 1.1 \mu\text{m}$, $h = 300 \text{ nm}$, and $g = 6.5 \mu\text{m}$. For the fabricated MZMs, the arm-length for phase-shifting is 5 mm. Correspondingly the metal absorption loss of 0.02 dB/cm, and the calculated voltage-length product $V_{\pi}L = 4.06 \text{ V}\cdot\text{cm}$ in theory. It should be noted that $V_{\pi}L$ can be reduced further by narrowing the gap width g if higher loss can be acceptable (e.g., 0.2 dB/cm).

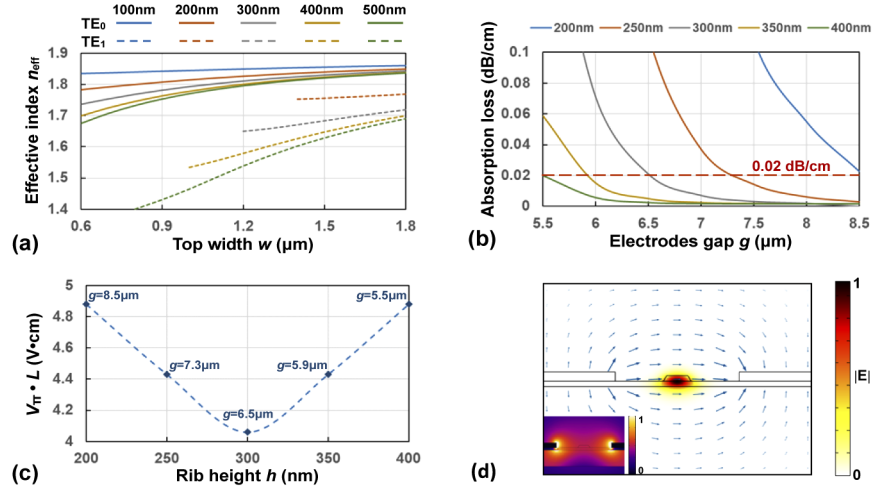


Fig. 2. (a) Calculated effective refractive indices for the TE₀ (solid) and TE₁ (dashed) modes in x-z plane waveguide of x-cut 600-nm LNOI platform at different rib heights h and top widths w ; (b) Calculated metal absorption of the LNOI waveguide when $w = 1.1 \mu\text{m}$; (c) Calculated voltage-length product $V_{\pi}L$; (d) Simulated mode profile $|E|$ of the TE₀ mode (colored) and electric field distribution (inset: the RF field profile at the frequency of $f = 100 \text{ GHz}$).

Figure 3(a) shows the top view of the designed 3-dB MMI coupler. The width and length of the MMI core are $W_{MMI} = 6 \mu\text{m}$ and $L_{MMI} = 22 \mu\text{m}$, respectively. The taper length l_{tp} and width w_{tp} of the input/output are $10 \mu\text{m}$ and $2 \mu\text{m}$, respectively. The separation between the output ports is $Y_{MMI} = 3.1 \mu\text{m}$. Figure 3(b) plots the simulated transmission at the upper- (solid) and lower- (dashed) ports, respectively, indicating uniform and low-loss power splitting in a wideband spectrum. The inset shows the simulated light propagation $|E|$ in the designed MMI coupler with the TE₀ mode input at $1.95 \mu\text{m}$.

The electrodes are designed with parameters of $g = 6.5 \mu\text{m}$, $t = 500 \text{ nm}$, $w_s = 17 \mu\text{m}$, and $w_g = 60 \mu\text{m}$ so as to satisfy the 50-Ω impedance-matching requirement. The calculated impedance $\text{Re}[Z_0]$ of the designed electrodes is shown in Fig. 4(a), indicating a good matching with 50 Ω (see the red dashed line). Figure 4(b) shows the calculated RF attenuation α_m , which is less than

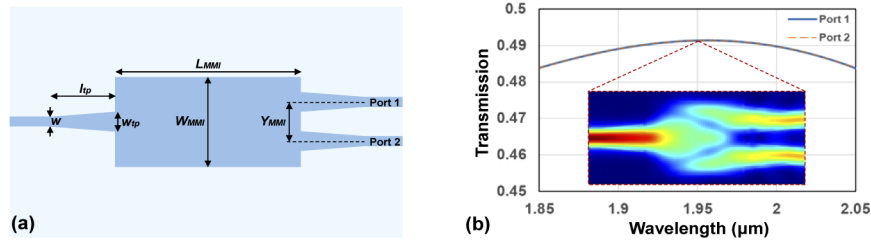


Fig. 3. (a) Top view of the designed 3-dB MMI coupler; (b) Calculated two-port transmissions of the designed MMI splitter for the TE₀-mode input, inset: simulated light propagation in the designed MMI at 1.95 μm.

9 dB/cm in the range of <100 GHz. Figure 4(c) shows the calculated effective index n_m of the RF mode. Here the group index of the optical mode is 2.276 (see the red dashed line). It can be seen that the velocity matching is also very good. In theory, the overall modulation frequency response $m[\omega]$ of a travelling wave modulator is given by [24]

$$m[\omega] = 20\log_{10} \left(\frac{2Z_{in}}{Z_{in} + 50} \cdot \frac{(50 + Z_0) \cdot F_+ + (50 - Z_0) \cdot F_-}{(50 + Z_0) \cdot e^{\gamma_m L} + (50 - Z_0) \cdot e^{-\gamma_m L}} \right), \quad (1)$$

where ω is the angular frequency of the RF signal, Z_{in} is the impedance seen at the entrance of the electrode, L is the length of the electrode, γ_m is the RF propagation constant. One has

$$Z_{in} = Z_0 \cdot \frac{50 + Z_0 \cdot \tanh(\gamma_m L)}{Z_0 + 50 \cdot \tanh(\gamma_m L)}, \quad (2)$$

$$\gamma_m = \alpha_m + j \frac{\omega}{c_o} n_o, \quad (3)$$

$$F_{\pm} = \frac{1 - e^{\pm \gamma_m L - j \frac{\omega}{c_o} n_o L}}{\pm \gamma_m L - j \frac{\omega}{c_o} n_o L}, \quad (4)$$

where n_o is the group index of the optical mode. In our calculation, the impedances of the source and load are assumed 50 Ω. Figure 4(d) shows the calculated modulation response $m[\omega]$ of the designed modulator. It can be seen that the theoretical 3-dB EO bandwidth far exceeds 100 GHz. Figure 4(e) shows the response of the first optical sideband with the open-circuit generator voltage of V_{π} , $V_{\pi}/2$, $V_{\pi}/4$, $V_{\pi}/8$ and $V_{\pi}/16$ [25], also showing >100 GHz EO bandwidth. And it can be seen the MZM has a good linearity in the range of 1~100GHz when the open-circuit generator voltage is less than $V_{\pi}/2$, which is consistent with the principle of linear electro-optic effect of LiNbO₃. Both the silica layer and silicon substrate are considered in these calculations.

The designed modulators were fabricated with a commercial *x*-cut LNOI wafer (NANOLN, China), in which a 600-nm LN thin film is bonded on top of a 2-μm-thick silica layer and a 0.4-mm-thick Si substrate. The fabrication processes were developed optimized carefully for achieving low-loss LNOI waveguides. Here the process of UV contact lithography (instead of E-beam lithography used previously) was used, which is helpful for the future massive production. The LN waveguide structure was etched physically by the Ar⁺ plasma etching process, in which way the waveguide has a sidewall angle of ~30°. And the electrodes are patterned through a secondary UV lithography process and a lift-off process. The photoresists used here are AZ MIR-701 and AZ 5214E, which are thick enough for etching and lift-off. Figure 5(a) shows the optical microscope image of our fabricated Mach-Zehnder modulators (MZMs), with the enlarged images of input waveguides, electrodes, as well as the power splitter. Figure 5(b) shows the scanning electron microscope (SEM) image of a butt-coupling end of the fabricated MZMs. It can be seen that the etched surface is very smooth, indicating that the LNOI waveguide loss is low.

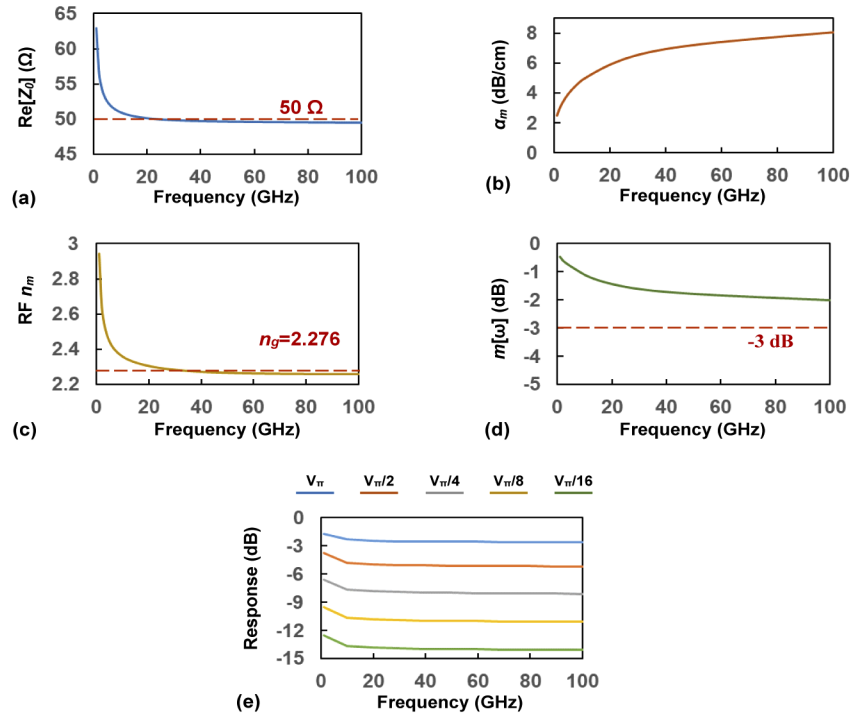


Fig. 4. Calculated results for the RF properties. (a) Impedance $\text{Re}[Z_0]$; (b) RF attenuation α_m ; (c) RF effective index n_m ; and (d) Calculated modulation response $m[\omega]$ of the designed modulator. Here the design is with $g = 6.5 \mu\text{m}$, $t = 500 \text{ nm}$, $w_s = 17 \mu\text{m}$, and $w_g = 60 \mu\text{m}$; (e) The calculated response of the first optical sideband with the open-circuit generator voltage of V_π , $V_\pi/2$, $V_\pi/4$, $V_\pi/8$, and $V_\pi/16$.

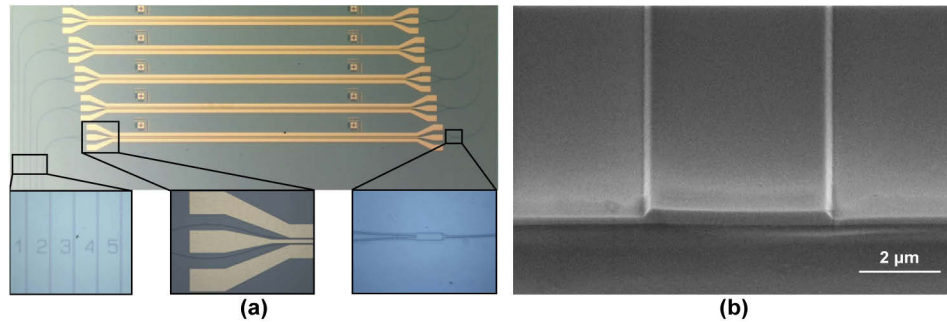


Fig. 5. (a) Optical microscope images of the fabricated MZMs, with the enlarged images of input waveguides, electrodes, and a power splitter; (b) An SEM image of a butt-coupling end of the fabricated MZMs.

3. Measurement results

The static EO response was measured with a 100-kHz triangular voltage sweep method. For the optical signal, light from a 1957 nm laser controlled by a polarization controller is butt-coupled into and out of the chip under test by using single-mode lensed fibers. For the present optical modulator, the butt-coupling loss is about 5 dB/facet, and the on-chip loss is about 6 dB, that is, the total insertion loss is about 16 dB. Figure 6(a) shows the normalized optical transmission as a function of the applied voltage. It can be seen that the half-wave voltage V_π is 7.34 V. The corresponding voltage-length product is $V_\pi L = 3.67$ V·cm, which is close to the theoretical value of 4.06 V·cm. The difference between the measured $V_\pi L$ and the theoretical result is partially due to some errors in the testing and the fabrication (e.g., the metal strips were widened due to the UV lithography). The inset of Fig. 6(a) shows the measured optical transmission on a logarithmic scale, indicating that the measured extinction ratio (ER) is > 20 dB.

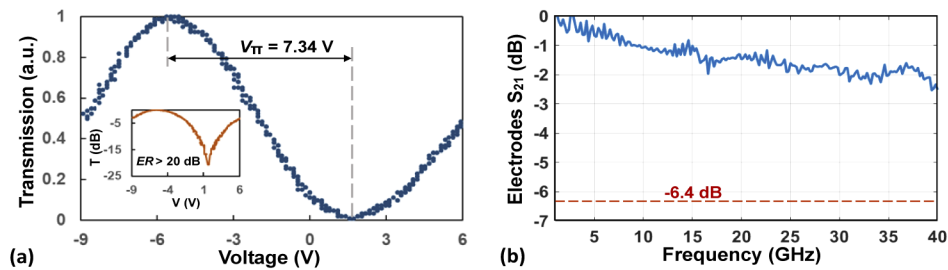


Fig. 6. (a) Normalized optical transmission as a function of the applied voltage; (b) Normalized S_{21} of the travelling-wave electrodes.

In the case of the velocity matching, the EO bandwidth of a modulator is mainly limited by the loss of the travelling-wave electrodes. The 3-dB EO bandwidth equals the frequency at which the total electrodes loss becomes 6.4 dB [26]. Figure 6(b) shows the measured S_{21} parameter of the travelling-wave electrodes. It can be seen that the measured 6.4 dB electric bandwidth far exceeds 40 GHz, which is beyond the measurement limits of the vector network analyzer (VNA) used here. The result was normalized with respect to the frequency response of the pair of high-speed ground-signal-ground (GSG) probes (0–40 GHz). From the frequency response trend, it can be predicted that the 6.4 dB bandwidth is around 100 GHz.

Figure 7(a) shows the measured bandwidth of the small-signal EO response (S_{21}) of the fabricated MZM. In the experiment, the RF signal was applied to the travelling-wave electrodes through a 40-GHz bandwidth RF probe (GGB 40A) and the RF signal after transmitting the device is terminated by attaching a second RF probe (GGB 40A) connected with a $50\ \Omega$ terminator to the electrodes. In our experiment the high-speed modulated signal was then received by a 2- μ m photodetector with a bandwidth of 18 GHz, which is the best one provided by the vendors. The result given in Fig. 7(a) was normalized with respect to the frequency response of the high-speed GSG probe (0–40 GHz) and the photodetector (0–22 GHz). Note that the photodetector response in the datasheet provided by the vendor is in the frequency range of < 22 GHz. From Fig. 7(a), it can be seen that the measured 3-dB EO bandwidth of the fabricated modulator is larger than 22 GHz. It should be noted the measured bandwidth here is limited by the photodetector instead of the MZM itself. According to the measured electrodes response in Fig. 6(b), it is reasonable to believe the EO bandwidth of the present MZM is far beyond 40 GHz.

Finally, the eye-diagrams of the on-off keying (OOK) modulation signals modulated by the fabricated Mach-Zehnder EO modulator were also measured experimentally. Here a pulse pattern generator (PPG) was used to generate the electrical signals, and a 35-GHz RF amplifier was used to amplify the driving signal, while a DC bias around the quadrature point. The received signal

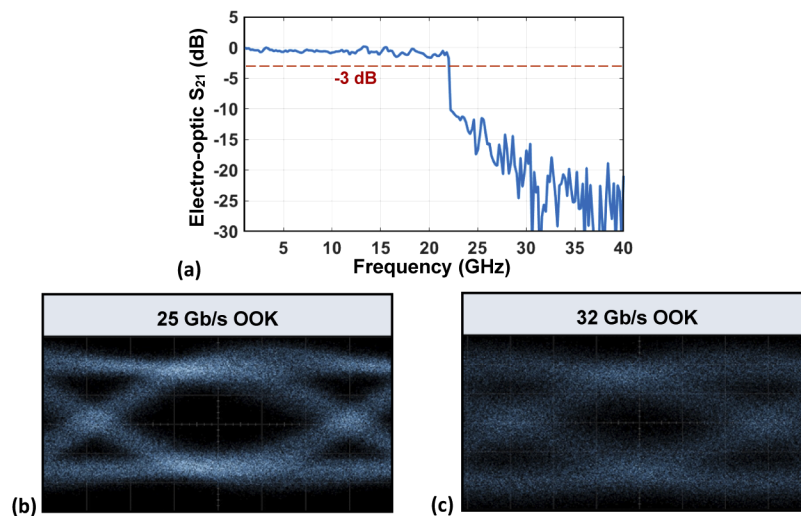


Fig. 7. (a) Normalized electro-optic S_{21} ; Eye-diagrams for the OOK signal at the data rates of 25 Gbps (b) and 32 Gbps (c).

is sampled by a wide-bandwidth oscilloscope (WBO). Figure 7(b)-(c) show the measured results for the signal with a bit rate of 25 Gb/s as well as 32 Gb/s, 32 Gb/s is the fastest data rate of the PPG in the experiment. In the experiment, the drive voltage is around 6 V, and correspondingly the energy efficiencies are estimated to be around 14 pJ/bit and 11 pJ/bit for the signals of 25 Gb/s and 32 Gb/s, respectively. From Fig. 7(b)-(c), it can be seen that the eyes are well open and the SNR (signal-to-noise ratio) for the 25 Gb/s signal is about 14 dB. In contrast, the eye quality for the 32 Gb/s signal is not as excellent as the 25 Gb/s signal because of the limited 3-dB bandwidth of the photodetector used in the experiment. Therefore, it is expected to achieve higher signal quality if a photodetector with a higher bandwidth is available in the future.

4. Conclusion

In conclusion, we have demonstrated the first EO MZMs working at the 2- μ m wavelength-band by using the LNOI platform. The presented device exhibits a voltage-length product of 3.67 V-cm, and an EO bandwidth of more than 22 GHz (which is limited by the measurement limit of photodetector). With more efforts in both fabrication and measurement, it is expected to achieve better performances of modulation in the aspects of insertion loss, modulation bandwidth, energy consumption, etc. Such an optimized EO MZM at 2- μ m is believed to play a key role in the applications of optical communication, optical sensing, and optical signal processing.

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